

Deborah Olofin

Data Scientist & Machine Learning Engineer · End-to-End ML Systems · Forecasting & Deployment

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PROFESSIONAL SUMMARY

Data Scientist and Machine Learning Engineer with a Computer Science foundation and a production ML system currently live and serving real users. Operates across the full spectrum from rigorous statistical analysis, time-series forecasting, and NLP through to end-to-end ML pipeline architecture and live deployment. Built a complete humanitarian forecasting system from raw multi-source data ingestion through to a deployed Streamlit application, independently delivering approximately 85-90% of the technical pipeline while leading a 12-person cross-functional team. Extends into deep learning, having built and trained feedforward neural networks and NLP sentiment classifiers using TensorFlow and Keras. Grounded in the mathematics, algorithms and systems thinking that sit underneath models, not just in running libraries. Communicates complex analytical findings clearly to both technical and non-technical stakeholders. Actively seeking remote Data Scientist or ML Engineer roles internationally where rigorous, end-to-end ownership and shipped systems matter.

TECHNICAL SKILLS

Languages & Libraries: Python · SQL · Pandas · NumPy · Scikit-learn · Matplotlib · Seaborn · TensorFlow · Keras

Machine Learning & Modelling: Time-Series Forecasting · Predictive Modelling · Regression · Clustering · Classification · Feature Engineering · Multi-Horizon Forecasting · Model Evaluation (MAPE/MAE/RMSE)

Deep Learning & NLP: Feedforward Neural Networks (FFNN) · Sentiment Analysis · NLP Text Classification · TensorFlow · Keras

Forecasting & ML Models: Facebook Prophet · XGBoost · SARIMA · SARIMAX · Naive Baseline Benchmarking · Cross-Validation · Multi-Seed Stability Testing

Pipeline & Data Engineering: End-to-End ML Pipeline Design · Multi-Source Data Integration · Data Cleaning & Preprocessing · Lag Feature Construction · Log Transformations · Structural Break Detection · EDA · Outlier Detection

Deployment & Tools: Streamlit · Git · GitHub · Version Control · Jupyter Notebook · Google Colab

Systems & CS Fundamentals: Algorithm Design · Data Structures · Software-Hardware Integration · Database Systems · Technical Documentation

Design & Communication: Data Storytelling · Stakeholder Communication · Graphic Design · Canva

PROJECTS

Early Warning Food Price Monitor — Northeast Nigeria · Team Lead · Lead Data Scientist · Primary ML Engineer · Oct 2025 – Mar 2026

Production-grade multi-model time-series forecasting system predicting monthly retail food prices up to 3 months ahead across conflict-affected states in Northeast Nigeria (Adamawa, Borno, Yobe). Independently responsible for approximately 85-90% of the complete technical pipeline spanning data sourcing, cleaning, feature engineering, all modelling, evaluation framework design, live deployment and full documentation while simultaneously leading a 12-person team across 4 sub-teams.

- ▶ **Data Engineering:** Sourced and cleaned 8 datasets from WFP, CBN, NBS, ACLED, NIMET and FAO. Engineered a master dataset of ~2,846 model-ready observations from 50,000+ raw rows spanning 13 states, 7 commodities and 8 years.
- ▶ **Feature Engineering:** Constructed lag variables (lag1/lag2/lag3), harvest season indicator, cross-commodity average price signal, subsidy removal structural break indicator and log transformations, all grounded in domain logic to prevent leakage.
- ▶ **Model Development:** Built and trained SARIMA, SARIMAX, XGBoost and Facebook Prophet. Evaluated all models against a naive baseline across 1-, 2- and 3-month forecast horizons (13 commodity-state combinations). Prophet selected as production model, winning 10 of 13 combinations.
- ▶ **Evaluation & Results:** Average MAPE: 30.8%. Best result: Rice (imported) Borno at 14.7% MAPE. 3 combinations rated Excellent (MAPE < 20%); 11 of 13 production-ready. All models beat naive baseline across all forecast horizons.
- ▶ **Deployment:** Designed and deployed the complete forecasting system as a live Streamlit web application, functional, accessible to non-technical humanitarian users, and currently running.

Live: earlywarningfoodpricemonitor-group3.streamlit.app · **GitHub:** github.com/Mi-kami/Early-Warning-Food-Price-Monitor-Northeast-Nigeria

Crypto AI Intelligence System · Personal Project · Jan 2026 – Present

End-to-end AI-powered cryptocurrency intelligence platform in active development. Designed and built a live multi-source data ingestion pipeline as Phase 1, with full error handling and happy-path testing complete. Currently in Phase 2: storage architecture and database design.

- ▶ Pipeline: Built and tested a production-grade data ingestion pipeline pulling from multiple live cryptocurrency data sources with full error handling and failure-mode coverage.
- ▶ Stack: Python, PostgreSQL, MLflow, Docker, Railway, Streamlit, CI/CD

NLP Sentiment Analysis System · Independent Project — Codveda Internship · Apr 2026

Built a complete NLP text classification pipeline to classify financial and social media sentiment into five categories. Applied full preprocessing pipeline including tokenisation, stopword removal and lemmatisation. Trained and evaluated multiple classifiers, achieving strong macro-F1 performance. Engineered features using TF-IDF vectorisation and validated results across stratified folds.

- ▶ Preprocessing: Built end-to-end text cleaning pipeline covering tokenisation, stopword removal, lemmatisation and noise filtering on a 732-row, 5-class sentiment dataset.
- ▶ Modelling: Trained Logistic Regression, Naive Bayes and Random Forest classifiers. Evaluated using macro-F1, precision, recall and confusion matrix analysis.
- ▶ Tools: Python, Pandas, NLTK, Scikit-learn, Matplotlib

Feedforward Neural Network — MNIST Image Classification · Independent Project — Codveda Internship · Apr 2026

Designed, built and trained a feedforward neural network from scratch using TensorFlow and Keras for handwritten digit classification on the MNIST dataset. Implemented full training pipeline including layer architecture design, activation functions, dropout regularisation, batch normalisation and learning rate tuning.

- ▶ Architecture: Multi-layer dense network with ReLU activations, dropout for regularisation, and softmax output layer for 10-class classification.
- ▶ Results: Achieved strong validation accuracy on MNIST test set. Tracked training and validation curves to diagnose overfitting and tune depth and dropout rates.
- ▶ Tools: Python, TensorFlow, Keras, NumPy, Matplotlib

Customer Segmentation System · Independent Project · 2025

- ▶ Built a K-means clustering pipeline for customer segmentation. Applied elbow method for optimal cluster selection, performed full EDA, and produced interpretable segment profiles to support downstream business decision-making.

Revenue Prediction — Data Scarcity Analysis · Independent Project · 2025

- ▶ Built a multiple linear regression pipeline on a deliberately constrained dataset (50 rows) to investigate model reliability under data scarcity. Documented train-test split sensitivity, overfitting patterns, and validated stability via cross-validation and multi-seed testing (variance: 0.139).

EXPERIENCE

Data Science Intern · Codveda Technologies (Remote) · Apr 2026

- ▶ Completed 6 end-to-end data science tasks spanning data cleaning, EDA, regression, classification, NLP sentiment analysis, and neural network design across structured, tabular and text datasets.
- ▶ Delivered production-quality Jupyter notebooks with full documentation, visualisations and analytical commentary for each task level.
- ▶ Tools: Python, Pandas, NumPy, Scikit-learn, TensorFlow, Keras, NLTK, Matplotlib, Seaborn

IT & Data Intern · Staco Insurance PLC · Nov 2023 – Mar 2024

- ▶ Worked with enterprise database systems with role-based access control in a financial services environment.
- ▶ Contributed to internal web development projects and gained exposure to software-hardware integration, networking fundamentals and enterprise system architecture.

Eco General Secretary · NACOS — OOU Chapter · 2023 – 2024

- ▶ Led organisational operations for the National Association of Computer Science Students, OOU chapter, coordinating activities, managing communications and representing student interests at institutional level.

EDUCATION & TRAINING

BSc Computer Science — Second Class Upper · Olabisi Onabanjo University · 2021 – 2024

CGPA: 3.78 / 5.00 · Core modules: Algorithms, Data Structures, Software Engineering, Database Systems, Computer Networks

Data Science Programme · TS Academy · Oct 2025 – Apr 2026

Completed structured data science curriculum. Capstone: production ML system deployed to live environment. Certificate received April 1, 2026. Credential ID: TSG2BTVKYE39LK

ML Engineering Career Track · DataCamp · Jan 2026 – Present

In progress — structured machine learning engineering programme. Currently on Course 5.

Applied Data Science Lab · WorldQuant University · Nov 2025 – Present

Rigorous applied ML programme. Currently on Project 4 of 8.

CERTIFICATIONS

- ▶ Data Science — TS Academy (April 2026) · Credential ID: TSG2BTVKYE39LK
- ▶ Data Science Mathematics — Duke University / Coursera (Nov 2025)
- ▶ Python Programming Intermediate — DataCamp (Nov 2025)
- ▶ Introduction to Python — DataCamp (Nov 2025)
- ▶ Soft Skills — Propel X Jobberman Nigeria